

Problem 7.6

Sally has two IRA's. IRA #1 earns interest at 8% effective annually and IRA #2 earns interest at 10% effective annually. She has not made any contributions since January 1, 1985, when the amount in IRA #1 was twice the amount in IRA #2. The sum of the two accounts on January 1, 1993 was \$75,000. Determine how much was in IRA #2 on January 1, 1985.

Solution

Let x be the amount in IRA #2 on January 1, 1985. Then the amount in IRA #1 on that date was $2x$.

From January 1, 1985 to January 1, 1993 is 8 years, so by January 1, 1993 the two accounts have grown to

$$2x(1.08)^8 \quad \text{and} \quad x(1.10)^8.$$

Since their sum is \$75,000,

$$75,000 = 2x(1.08)^8 + x(1.10)^8.$$

Factoring out x ,

$$75,000 = x(2(1.08)^8 + (1.10)^8).$$

Using

$$(1.08)^8 \approx 1.850930, \quad (1.10)^8 \approx 2.143589,$$

we get

$$75,000 \approx x(3.701860 + 2.143589) = 5.845449x.$$

Therefore,

$$x = \frac{75,000}{5.845449} \approx 12,830.49.$$

So the amount in IRA #2 on January 1, 1985 was

$$\boxed{\$12,830.49}.$$